

Model for “0h1”

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General Condition

- Basis type: **lgs**
- SAMB selection:
 - Type: [Q, G]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , E_g , E_u , T_{1g} , T_{1u} , T_{2g} , T_{2u}]
 - Spin (s): [0, 1]
- Atomic selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , E_g , E_u , T_{1g} , T_{1u} , T_{2g} , T_{2u}]
 - Spin (s): [0, 1]
- Site-cluster selection:
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , E_g , E_u , T_{1g} , T_{1u} , T_{2g} , T_{2u}]
- Bond-cluster selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_{1g} , A_{1u} , A_{2g} , A_{2u} , E_g , E_u , T_{1g} , T_{1u} , T_{2g} , T_{2u}]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: **false**

Group and Unit Cell

- Group: SG No. 221 $O_h^1 Pm\bar{3}m$ [cubic]
- Associated point group: PG No. 221 $O_h m\bar{3}m$ [cubic]
- Unit cell:
 - $a = 1.00000$, $b = 1.00000$, $c = 1.00000$, $\alpha = 90.0$, $\beta = 90.0$, $\gamma = 90.0$
- Lattice vectors (conventional cell):
 - $\mathbf{a}_1 = [1.00000, 0.00000, 0.00000]$
 - $\mathbf{a}_2 = [0.00000, 1.00000, 0.00000]$
 - $\mathbf{a}_3 = [0.00000, 0.00000, 1.00000]$

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1 0\}$	2	$\{2_{001} 0\}$	3	$\{2_{010} 0\}$	4	$\{2_{100} 0\}$	5	$\{3_{111}^+ 0\}$
6	$\{3_{-11-1}^+ 0\}$	7	$\{3_{1-1-1}^+ 0\}$	8	$\{3_{-1-11}^+ 0\}$	9	$\{3_{111}^- 0\}$	10	$\{3_{1-1-1}^- 0\}$
11	$\{3_{-1-11}^- 0\}$	12	$\{3_{-11-1}^- 0\}$	13	$\{2_{110} 0\}$	14	$\{2_{1-10} 0\}$	15	$\{4_{001}^- 0\}$
16	$\{4_{001}^+ 0\}$	17	$\{4_{100}^- 0\}$	18	$\{2_{011} 0\}$	19	$\{2_{01-1} 0\}$	20	$\{4_{100}^+ 0\}$
21	$\{4_{010}^+ 0\}$	22	$\{2_{101} 0\}$	23	$\{4_{010}^- 0\}$	24	$\{2_{-101} 0\}$	25	$\{-1 0\}$
26	$\{m_{001} 0\}$	27	$\{m_{010} 0\}$	28	$\{m_{100} 0\}$	29	$\{-3_{111}^+ 0\}$	30	$\{-3_{-11-1}^+ 0\}$
31	$\{-3_{1-1-1}^+ 0\}$	32	$\{-3_{-1-11}^+ 0\}$	33	$\{-3_{111}^- 0\}$	34	$\{-3_{1-1-1}^- 0\}$	35	$\{-3_{-1-11}^- 0\}$
36	$\{-3_{-11-1}^- 0\}$	37	$\{m_{110} 0\}$	38	$\{m_{1-10} 0\}$	39	$\{-4_{001}^- 0\}$	40	$\{-4_{001}^+ 0\}$
41	$\{-4_{100}^- 0\}$	42	$\{m_{011} 0\}$	43	$\{m_{01-1} 0\}$	44	$\{-4_{100}^+ 0\}$	45	$\{-4_{010}^+ 0\}$
46	$\{m_{101} 0\}$	47	$\{-4_{010}^- 0\}$	48	$\{m_{-101} 0\}$				

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_{1g})$	A_{1g}	0	Q, T	-	-	1

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
2	$\mathbb{Q}_4(A_{1g})$	A_{1g}	4	Q, T	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$
3	$\mathbb{G}_0(A_{1u})$	A_{1u}	0	G, M	-	-	1
4	$\mathbb{G}_4(A_{1u})$	A_{1u}	4	G, M	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$
5	$\mathbb{G}_3(A_{2g})$	A_{2g}	3	G, M	-	-	$\sqrt{15}xyz$
6	$\mathbb{Q}_3(A_{2u})$	A_{2u}	3	Q, T	-	-	$\sqrt{15}xyz$
7	$\mathbb{Q}_{2,1}(E_g)$	E_g	2	Q, T	-	1	$\frac{-x^2-y^2+2z^2}{2}$
8	$\mathbb{Q}_{2,2}(E_g)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
9	$\mathbb{Q}_{4,1}(E_g)$	E_g	4	Q, T	-	1	$\frac{\sqrt{15}(-x^4+12x^2y^2-6x^2z^2-y^4-6y^2z^2+2z^4)}{12}$
10	$\mathbb{Q}_{4,2}(E_g)$					2	$\frac{\sqrt{5}(x^4-6x^2z^2-y^4+6y^2z^2)}{4}$
11	$\mathbb{G}_{2,1}(E_u)$	E_u	2	G, M	-	1	$\frac{-x^2-y^2+2z^2}{2}$
12	$\mathbb{G}_{2,2}(E_u)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
13	$\mathbb{G}_{4,1}(E_u)$	E_u	4	G, M	-	1	$\frac{\sqrt{15}(-x^4+12x^2y^2-6x^2z^2-y^4-6y^2z^2+2z^4)}{12}$
14	$\mathbb{G}_{4,2}(E_u)$					2	$\frac{\sqrt{5}(x^4-6x^2z^2-y^4+6y^2z^2)}{4}$
15	$\mathbb{Q}_{5,1}(E_u)$	E_u	5	Q, T	-	1	$\frac{3\sqrt{35}xyz(x^2-y^2)}{2}$
16	$\mathbb{Q}_{5,2}(E_u)$					2	$\frac{\sqrt{105}xyz(x^2+y^2-2z^2)}{2}$
17	$\mathbb{G}_{1,1}(T_{1g})$	T_{1g}	1	G, M	-	1	x
18	$\mathbb{G}_{1,2}(T_{1g})$					2	y
19	$\mathbb{G}_{1,3}(T_{1g})$					3	z
20	$\mathbb{G}_{3,1}(T_{1g})$	T_{1g}	3	G, M	-	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
21	$\mathbb{G}_{3,2}(T_{1g})$					2	$\frac{y(-3x^2+2y^2-3z^2)}{2}$
22	$\mathbb{G}_{3,3}(T_{1g})$					3	$\frac{z(-3x^2-3y^2+2z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
23	$\mathbb{Q}_{4,1}(T_{1g})$	T_{1g}	4	Q, T	-	1	$\frac{\sqrt{35}yz(y^2-z^2)}{2}$
24	$\mathbb{Q}_{4,2}(T_{1g})$					2	$\frac{\sqrt{35}xz(-x^2+z^2)}{2}$
25	$\mathbb{Q}_{4,3}(T_{1g})$					3	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
26	$\mathbb{Q}_{1,1}(T_{1u})$	T_{1u}	1	Q, T	-	1	x
27	$\mathbb{Q}_{1,2}(T_{1u})$					2	y
28	$\mathbb{Q}_{1,3}(T_{1u})$					3	z
29	$\mathbb{Q}_{3,1}(T_{1u})$	T_{1u}	3	Q, T	-	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
30	$\mathbb{Q}_{3,2}(T_{1u})$					2	$\frac{y(-3x^2+2y^2-3z^2)}{2}$
31	$\mathbb{Q}_{3,3}(T_{1u})$					3	$\frac{z(-3x^2-3y^2+2z^2)}{2}$
32	$\mathbb{G}_{4,1}(T_{1u})$	T_{1u}	4	G, M	-	1	$\frac{\sqrt{35}yz(y^2-z^2)}{2}$
33	$\mathbb{G}_{4,2}(T_{1u})$					2	$\frac{\sqrt{35}xz(-x^2+z^2)}{2}$
34	$\mathbb{G}_{4,3}(T_{1u})$					3	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
35	$\mathbb{Q}_{5,1}(T_{1u}, 2)$	T_{1u}	5	Q, T	2	1	$\frac{3\sqrt{35}x(y^4-6y^2z^2+z^4)}{8}$
36	$\mathbb{Q}_{5,2}(T_{1u}, 2)$					2	$\frac{3\sqrt{35}y(x^4-6x^2z^2+z^4)}{8}$
37	$\mathbb{Q}_{5,3}(T_{1u}, 2)$					3	$\frac{3\sqrt{35}z(x^4-6x^2y^2+y^4)}{8}$
38	$\mathbb{Q}_{2,1}(T_{2g})$	T_{2g}	2	Q, T	-	1	$\sqrt{3}yz$
39	$\mathbb{Q}_{2,2}(T_{2g})$					2	$\sqrt{3}xz$
40	$\mathbb{Q}_{2,3}(T_{2g})$					3	$\sqrt{3}xy$
41	$\mathbb{G}_{3,1}(T_{2g})$	T_{2g}	3	G, M	-	1	$\frac{\sqrt{15}x(y^2-z^2)}{2}$
42	$\mathbb{G}_{3,2}(T_{2g})$					2	$\frac{\sqrt{15}y(-x^2+z^2)}{2}$
43	$\mathbb{G}_{3,3}(T_{2g})$					3	$\frac{\sqrt{15}z(x^2-y^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
44	$\mathbb{Q}_{4,1}(T_{2g})$	T_{2g}	4	Q, T	-	1	$\frac{\sqrt{5}yz(6x^2-y^2-z^2)}{2}$
45	$\mathbb{Q}_{4,2}(T_{2g})$					2	$\frac{\sqrt{5}xz(-x^2+6y^2-z^2)}{2}$
46	$\mathbb{Q}_{4,3}(T_{2g})$					3	$\frac{\sqrt{5}xy(-x^2-y^2+6z^2)}{2}$
47	$\mathbb{G}_{2,1}(T_{2u})$	T_{2u}	2	G, M	-	1	$\sqrt{3}yz$
48	$\mathbb{G}_{2,2}(T_{2u})$					2	$\sqrt{3}xz$
49	$\mathbb{G}_{2,3}(T_{2u})$					3	$\sqrt{3}xy$
50	$\mathbb{Q}_{3,1}(T_{2u})$	T_{2u}	3	Q, T	-	1	$\frac{\sqrt{15}x(y^2-z^2)}{2}$
51	$\mathbb{Q}_{3,2}(T_{2u})$					2	$\frac{\sqrt{15}y(-x^2+z^2)}{2}$
52	$\mathbb{Q}_{3,3}(T_{2u})$					3	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
53	$\mathbb{G}_{4,1}(T_{2u})$	T_{2u}	4	G, M	-	1	$\frac{\sqrt{5}yz(6x^2-y^2-z^2)}{2}$
54	$\mathbb{G}_{4,2}(T_{2u})$					2	$\frac{\sqrt{5}xz(-x^2+6y^2-z^2)}{2}$
55	$\mathbb{G}_{4,3}(T_{2u})$					3	$\frac{\sqrt{5}xy(-x^2-y^2+6z^2)}{2}$
56	$\mathbb{Q}_{5,1}(T_{2u})$	T_{2u}	5	Q, T	-	1	$\frac{\sqrt{105}x(2x^2y^2-2x^2z^2-y^4+z^4)}{4}$
57	$\mathbb{Q}_{5,2}(T_{2u})$					2	$\frac{\sqrt{105}y(x^4-2x^2y^2+2y^2z^2-z^4)}{4}$
58	$\mathbb{Q}_{5,3}(T_{2u})$					3	$\frac{\sqrt{105}z(-x^4+2x^2z^2+y^4-2y^2z^2)}{4}$

 Basis in full matrix

Table 3: dimension = 8

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$ s, \uparrow\rangle @A(1)$	1	$ s, \downarrow\rangle @A(1)$	2	$ p_x, \uparrow\rangle @A(1)$	3	$ p_x, \downarrow\rangle @A(1)$	4	$ p_y, \uparrow\rangle @A(1)$
5	$ p_y, \downarrow\rangle @A(1)$	6	$ p_z, \uparrow\rangle @A(1)$	7	$ p_z, \downarrow\rangle @A(1)$				

Table 4: Atomic basis (orbital part only)

orbital	definition
$ s\rangle$	1
$ p_x\rangle$	x
$ p_y\rangle$	y
$ p_z\rangle$	z

— SAMB: 604 (all 604) —

- A : 'A' site-cluster

- * bra: $\langle s, \uparrow|, \langle s, \downarrow|$

- * ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$

- * wyckoff: **1a**

$$\boxed{\text{z1}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

- A : 'A' site-cluster

- * bra: $\langle s, \uparrow|, \langle s, \downarrow|$

* ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

* wyckoff: **1a**

$$\boxed{\text{z21}} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}) = \mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z117}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z118}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z263}} \quad \mathbb{Q}_{1,1}^{(c)}(T_{1u}) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z264}} \quad \mathbb{Q}_{1,2}^{(c)}(T_{1u}) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z265}} \quad \mathbb{Q}_{1,3}^{(c)}(T_{1u}) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z266}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_{1u}) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z267}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_{1u}) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z268}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_{1u}) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z485}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z486}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z487}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

- A : 'A' site-cluster

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$
 * ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$
 * wyckoff: **1a**

$$\boxed{\text{z2}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z3}} \quad \mathbb{Q}_0^{(1,1;c)}(A_{1g}) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z61}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z62}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z63}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z64}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$$

$$\boxed{\text{z179}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_{1g}) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z180}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_{1g}) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z181}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_{1g}) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z386}} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z387}} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z388}} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z389}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_{2g}) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z390}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_{2g}) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

$$\boxed{\text{z391}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_{2g}) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(s)}(A_{1g})}{3}$$

• **A;A_001_1** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$

* wyckoff: **3a03d**

$$\boxed{\text{z4}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z22}} \quad \mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z65}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z66}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z119}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z120}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z269}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z270}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z271}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z488}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z489}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z490}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

• **A;A_001_1** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

* wyckoff: **3a@3d**

$$\boxed{\text{z5}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{1,2}^{(a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z6}} \quad \mathbb{Q}_0^{(1,0;c)}(A_{1g}) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z23}} \quad \mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z24}} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}) = \mathbb{G}_0^{(1,1;a)}(A_{1u}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z39}} \quad \mathbb{G}_3^{(1,-1;c)}(A_{2g}) = \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z48}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z67}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{T}_{1,3}^{(a)}(T_{1u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z68}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z69}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z70}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z171}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z172}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z121}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z122}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z123}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z124}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z125}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z126}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z182}} \quad \mathbb{G}_{1,1}^{(c)}(T_{1g}) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z183}} \quad \mathbb{G}_{1,2}^{(c)}(T_{1g}) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z184}} \quad \mathbb{G}_{1,3}^{(c)}(T_{1g}) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z185}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{30}$$

$$\boxed{\text{z186}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30}$$

$$\begin{aligned}
\text{z187} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_{1g}) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30} \\
\text{z188} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_{1g}) &= \frac{\sqrt{5} \left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{30} \\
\text{z189} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_{1g}) &= -\frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30} \\
\text{z190} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_{1g}) &= \frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{15} \\
\text{z191} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_{1g}) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z192} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_{1g}) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z193} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_{1g}) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z194} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{3} \\
\text{z195} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{3} \\
\text{z196} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u})}{3} \\
\text{z272} \quad \mathbb{Q}_{1,1}^{(c)}(T_{1u}, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\text{z273} \quad \mathbb{Q}_{1,2}^{(c)}(T_{1u}, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\text{z274} \quad \mathbb{Q}_{1,3}^{(c)}(T_{1u}, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}
\end{aligned}$$

$$\boxed{\text{z275}} \quad \mathbb{Q}_{1,1}^{(c)}(T_{1u}, b) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z276}} \quad \mathbb{Q}_{1,2}^{(c)}(T_{1u}, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z277}} \quad \mathbb{Q}_{1,3}^{(c)}(T_{1u}, b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z278}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}) = \frac{3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z279}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}) = \frac{-3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z280}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}) = -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z281}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_{1u}, a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z282}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_{1u}, a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z283}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_{1u}, a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z284}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_{1u}, b) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z285}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_{1u}, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z286}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_{1u}, b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z392}} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\begin{aligned}
\text{z393} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z394} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{1,2}^{(a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z395} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z396} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z397} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2 \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z398} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_{2g}) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z399} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_{2g}) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z400} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_{2g}) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z401} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_{2g}) &= \frac{-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - 2 \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + 2 \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,2}^{(b)}(T_{1u})}{6} \\
\text{z402} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2 \mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2 \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,1}^{(b)}(T_{1u})}{6} \\
\text{z403} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_{2g}) &= \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,1}^{(b)}(T_{1u})}{3} \\
\text{z491} \quad \mathbb{Q}_{3,1}^{(c)}(T_{2u}) &= -\frac{3 \mathbb{Q}_{1,1}^{(a)}(T_{1u}) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\text{z492} \quad \mathbb{Q}_{3,2}^{(c)}(T_{2u}) &= \frac{3 \mathbb{Q}_{1,2}^{(a)}(T_{1u}) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}
\end{aligned}$$

$$\boxed{\text{z493}} \quad \mathbb{Q}_{3,3}^{(c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z494}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{2u}) = \frac{-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z495}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{2u}) = -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z496}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z497}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{2u}) = -\frac{3\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z498}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{2u}) = \frac{3\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z499}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z500}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z501}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z502}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

• **A;A_001_1** : 'A'-'A' bond-cluster

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: **3a@3d**

$$\boxed{\text{z7}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z8}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z9}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_{1g}) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z10}} \quad \mathbb{Q}_0^{(1,1;c)}(A_{1g}) = \mathbb{Q}_0^{(1,1;a)}(A_{1g}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z25}} \quad \mathbb{G}_0^{(c)}(A_{1u}) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z26}} \quad \mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z27}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z28}} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z40}} \quad \mathbb{G}_3^{(c)}(A_{2g}) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z41}} \quad \mathbb{G}_3^{(1,-1;c)}(A_{2g}) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z49}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = - \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z50}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}) = \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z73}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z74}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z75}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z76}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z77}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z78}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z79}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z80}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z81}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z82}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z83}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z84}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z127}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u})\right)}{6}$$

$$\boxed{\text{z128}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z129}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u})\right)}{28}$$

$$\boxed{\text{z130}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14}\left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u})\right)}{84}$$

$$\begin{aligned}
\text{z131} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) &= \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6} \\
\text{z132} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) &= \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2} \\
\text{z133} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u) &= \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{84} \\
\text{z134} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u) &= \frac{\sqrt{14} \left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{28} \\
\text{z135} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2} \\
\text{z136} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u) &= \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6} \\
\text{z137} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) &= \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6} \\
\text{z138} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) &= \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2} \\
\text{z197} \quad \mathbb{Q}_{4,1}^{(c)}(T_{1g}) &= -\frac{3\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\text{z198} \quad \mathbb{Q}_{4,2}^{(c)}(T_{1g}) &= \frac{3\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\text{z199} \quad \mathbb{Q}_{4,3}^{(c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\text{z200} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_{1g}) &= -\frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\text{z201} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_{1g}) &= \frac{3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z202}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z203}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_{1g},a) &= \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z204}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_{1g},a) &= \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z205}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_{1g},a) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z206}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_{1g},b) &= \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z207}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_{1g},b) &= -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z208}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_{1g},b) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z287}} \quad \mathbb{Q}_{1,1}^{(c)}(T_{1u}) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u})\right)}{6} \\
\boxed{\text{z288}} \quad \mathbb{Q}_{1,2}^{(c)}(T_{1u}) &= \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u})\right)}{6} \\
\boxed{\text{z289}} \quad \mathbb{Q}_{1,3}^{(c)}(T_{1u}) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u})\right)}{6} \\
\boxed{\text{z290}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u})\right)}{6} \\
\boxed{\text{z291}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}) &= \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u})\right)}{6} \\
\boxed{\text{z292}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u})\right)}{6}
\end{aligned}$$

$$\boxed{\text{z293}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{1u}) = \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z294}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{1u}) = \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z295}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{1u}) = \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z296}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_{1u}) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{30}$$

$$\boxed{\text{z297}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_{1u}) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30}$$

$$\boxed{\text{z298}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_{1u}) = \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30}$$

$$\boxed{\text{z299}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{1u}) = \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{30}$$

$$\boxed{\text{z300}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{1u}) = -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30}$$

$$\boxed{\text{z301}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{1u}) = \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{15}$$

$$\boxed{\text{z302}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_{1u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z303}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_{1u}) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z304}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_{1u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z305}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z306}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_{1u}) = \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z307}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z404}} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z405}} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z406}} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z407}} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z408}} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z409}} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z410}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z411}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z412}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z413}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_{2g}, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z414}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_{2g}, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z415}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_{2g}, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z416}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_{2g}) = -\frac{3\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z417}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_{2g}) = \frac{3\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z418}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_{2g}) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z503}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 4\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z504}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 4\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z505}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 4\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z506}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{2u}) = \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{6}$$

$$\boxed{\text{z507}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{2u}) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{6}$$

$$\boxed{\text{z508}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{2u}) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{3}$$

$$\boxed{\text{z509}} \quad \mathbb{G}_{2,1}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z510}} \quad \mathbb{G}_{2,2}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z511}} \quad \mathbb{G}_{2,3}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z512}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}, a) = \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{21}$$

$$\begin{aligned}
\boxed{\text{z513}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}, a) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{21} \\
\boxed{\text{z514}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}, a) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{21} \\
\boxed{\text{z515}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z516}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z517}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z518}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 12\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{84} \\
\boxed{\text{z519}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 12\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{84} \\
\boxed{\text{z520}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 12\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{84} \\
\boxed{\text{z521}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_{2u}) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z522}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_{2u}) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z523}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_{2u}) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z524}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z525}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z526}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

• **A;A_002_1** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$

* wyckoff: **6b03c**

$$\boxed{\text{z11}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z29}} \quad \mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z51}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3}$$

$$\boxed{\text{z85}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z86}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z139}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z140}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z141}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u})}{2}$$

$$\boxed{\text{z142}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z308}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}, a) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z309}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}, a) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\begin{aligned}
\text{z310} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z311} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6} \\
\text{z312} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z313} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z419} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}) &= \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(T_{2g})}{3} \\
\text{z420} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}) &= \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g})}{3} \\
\text{z421} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}) &= \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g})}{3} \\
\text{z527} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6} \\
\text{z528} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z529} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z530} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z531} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z532} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}
\end{aligned}$$

• **A;A_002_1** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

* wyckoff: **6b03c**

$$\boxed{\text{z12}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{1,2}^{(a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z13}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_{1g}) = \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3}$$

$$\boxed{\text{z14}} \quad \mathbb{Q}_0^{(1,0;c)}(A_{1g}) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z30}} \quad \mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{5} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{5}$$

$$\boxed{\text{z31}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{30} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z32}} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}) = \mathbb{G}_0^{(1,1;a)}(A_{1u}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z42}} \quad \mathbb{G}_3^{(c)}(A_{2g}) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{T}_{1,2}^{(a)}(T_{1u}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3}$$

$$\boxed{\text{z43}} \quad \mathbb{G}_3^{(1,-1;c)}(A_{2g}) = \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z44}} \quad \mathbb{G}_3^{(1,0;c)}(A_{2g}) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3}$$

$$\boxed{\text{z52}} \quad \mathbb{Q}_3^{(c)}(A_{2u}) = \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(a)}(T_{1u}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{1,2}^{(a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{Q}_{1,3}^{(a)}(T_{1u}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{3}$$

$$\boxed{\text{z53}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z54}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}) = \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{3}$$

$$\boxed{\text{z87}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z88}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z89}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u})}{2}$$

$$\boxed{\text{z90}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z91}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z92}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z93}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z94}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,2}^{(b)}(T_{2u})}{2}$$

$$\boxed{\text{z95}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, a) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z96}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, a) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z97}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, b) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u})}{2}$$

$$\boxed{\text{z98}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, b) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z143}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g})}{2}$$

$$\begin{aligned}
\text{z144} \quad \mathbb{G}_{2,2}^{(c)}(E_u) &= \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{6} \\
\text{z145} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) &= \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\text{z146} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) &= \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2} \\
\text{z147} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) &= \frac{\sqrt{7} \left(2\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{14} \\
\text{z148} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) &= \frac{\sqrt{7} \left(-2\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{14} \\
\text{z149} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u) &= \frac{\sqrt{21} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{42} \\
\text{z150} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{14} \\
\text{z151} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u) &= \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g})}{2} \\
\text{z152} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u) &= \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{6} \\
\text{z153} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) &= \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} \\
\text{z154} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) &= \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} \\
\text{z209} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_{1g}) &= -\frac{3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{2,1}^{(b)}(T_{2u})}{6} \\
\text{z210} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_{1g}) &= \frac{3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{2,2}^{(b)}(T_{2u})}{6}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z211}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{2,3}^{(b)}(T_{2u})}{3} \\
\boxed{\text{z212}} \quad \mathbb{G}_{1,1}^{(c)}(T_{1g},a) &= \frac{\sqrt{6}\left(\mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u})\right)}{6} \\
\boxed{\text{z213}} \quad \mathbb{G}_{1,2}^{(c)}(T_{1g},a) &= \frac{\sqrt{6}\left(-\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u})\right)}{6} \\
\boxed{\text{z214}} \quad \mathbb{G}_{1,3}^{(c)}(T_{1g},a) &= \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u})\right)}{6} \\
\boxed{\text{z215}} \quad \mathbb{G}_{1,1}^{(c)}(T_{1g},b) &= \frac{\sqrt{6}\left(\mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u})\right)}{6} \\
\boxed{\text{z216}} \quad \mathbb{G}_{1,2}^{(c)}(T_{1g},b) &= \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u})\right)}{6} \\
\boxed{\text{z217}} \quad \mathbb{G}_{1,3}^{(c)}(T_{1g},b) &= \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u})\right)}{6} \\
\boxed{\text{z218}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_{1g},a) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u})\right)}{30} \\
\boxed{\text{z219}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_{1g},a) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u})\right)}{30} \\
\boxed{\text{z220}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_{1g},a) &= \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u})\right)}{30} \\
\boxed{\text{z221}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_{1g},b) &= \frac{\sqrt{6}\left(-\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,2}^{(b)}(T_{2u})\right)}{6} \\
\boxed{\text{z222}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_{1g},b) &= \frac{\sqrt{6}\left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,1}^{(b)}(T_{2u})\right)}{6} \\
\boxed{\text{z223}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_{1g},b) &= \frac{\sqrt{6}\left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,1}^{(b)}(T_{2u})\right)}{6}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z224}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_{1g}) &= \frac{\sqrt{5} \left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{30} \\
\boxed{\text{z225}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_{1g}) &= -\frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30} \\
\boxed{\text{z226}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_{1g}) &= \frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{15} \\
\boxed{\text{z227}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_{1g}, a) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z228}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_{1g}, a) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z229}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_{1g}, a) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z230}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_{1g}, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6} \\
\boxed{\text{z231}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_{1g}, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\boxed{\text{z232}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_{1g}, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\boxed{\text{z233}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{3} \\
\boxed{\text{z234}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{3} \\
\boxed{\text{z235}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_{1g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u})}{3} \\
\boxed{\text{z314}} \quad \mathbb{Q}_{1,1}^{(c)}(T_{1u}, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}
\end{aligned}$$

$$\boxed{\text{z315}} \quad \mathbb{Q}_{1,2}^{(c)}(T_{1u}, a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z316}} \quad \mathbb{Q}_{1,3}^{(c)}(T_{1u}, a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z317}} \quad \mathbb{Q}_{1,1}^{(c)}(T_{1u}, b) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z318}} \quad \mathbb{Q}_{1,2}^{(c)}(T_{1u}, b) = \frac{\sqrt{10} \left(3\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z319}} \quad \mathbb{Q}_{1,3}^{(c)}(T_{1u}, b) = \frac{\sqrt{10} \left(3\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z320}} \quad \mathbb{Q}_{3,1}^{(c)}(T_{1u}) = \frac{\sqrt{5} \left(-3\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 2\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z321}} \quad \mathbb{Q}_{3,2}^{(c)}(T_{1u}) = -\frac{\sqrt{5} \left(2\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z322}} \quad \mathbb{Q}_{3,3}^{(c)}(T_{1u}) = \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{15}$$

$$\boxed{\text{z323}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{10} \left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z324}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{10} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z325}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{30} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 2\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z326}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{10} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 4\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{60}$$

$$\begin{aligned}
\text{z327} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{1u}) &= \frac{\sqrt{10} \left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) - 4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{60} \\
\text{z328} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{1u}) &= \frac{\sqrt{30} \left(-2\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30} \\
\text{z329} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_{1u}, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\text{z330} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_{1u}, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\text{z331} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_{1u}, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\text{z332} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_{1u}, b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30} \\
\text{z333} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_{1u}, b) &= \frac{\sqrt{10} \left(3\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30} \\
\text{z334} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_{1u}, b) &= \frac{\sqrt{10} \left(3\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{30} \\
\text{z335} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{1u}) &= \frac{\sqrt{5} \left(-3\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 2\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30} \\
\text{z336} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{1u}) &= -\frac{\sqrt{5} \left(2\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30} \\
\text{z337} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{1u}) &= \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{15} \\
\text{z338} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_{1u}) &= -\frac{\sqrt{2} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{12}
\end{aligned}$$

$$\begin{aligned}
\text{z339} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_{1u}) &= \frac{\sqrt{2} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12} \\
\text{z340} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_{1u}) &= \frac{\sqrt{6} \left(\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6} \\
\text{z422} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}, a) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z423} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}, a) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z424} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}, a) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z425} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}, b) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6} \\
\text{z426} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \mathbb{T}_{1,3}^{(a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z427} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}, b) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{1,2}^{(a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z428} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_{2g}, a) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z429} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_{2g}, a) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z430} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_{2g}, a) &= \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z431} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_{2g}, b) &= \frac{\sqrt{10} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{2,1}^{(b)}(T_{2u}) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{2,1}^{(b)}(T_{2u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{30} \\
\text{z432} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_{2g}, b) &= \frac{\sqrt{10} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{30}
\end{aligned}$$

$$\begin{aligned}
\text{z433} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_{2g}, b) &= \frac{\sqrt{10} \left(-2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{2,3}^{(b)}(T_{2u}) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{30} \\
\text{z434} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{5} \left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{2,1}^{(b)}(T_{2u}) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{2,1}^{(b)}(T_{2u}) + 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{30} \\
\text{z435} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{5} \left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{30} \\
\text{z436} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{15} \\
\text{z437} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_{2g}, a) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z438} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_{2g}, a) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z439} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_{2g}, a) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z440} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_{2g}, b) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6} \\
\text{z441} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_{2g}, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z442} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_{2g}, b) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z443} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_{2g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{M}_{2,1}^{(b)}(T_{2u})}{3} \\
\text{z444} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_{2g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{M}_{2,2}^{(b)}(T_{2u})}{3} \\
\text{z445} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_{2g}) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{M}_{2,3}^{(b)}(T_{2u})}{3}
\end{aligned}$$

$$\boxed{\text{z446}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_{2g}) = \frac{-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{6}$$

$$\boxed{\text{z447}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_{2g}) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{6}$$

$$\boxed{\text{z448}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_{2g}) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{3}$$

$$\boxed{\text{z533}} \quad \mathbb{Q}_{3,1}^{(c)}(T_{2u}) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 2\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g})}{6}$$

$$\boxed{\text{z534}} \quad \mathbb{Q}_{3,2}^{(c)}(T_{2u}) = \frac{-2\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g})}{6}$$

$$\boxed{\text{z535}} \quad \mathbb{Q}_{3,3}^{(c)}(T_{2u}) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z536}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g})\right)}{12}$$

$$\boxed{\text{z537}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{2}\left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g)\right)}{12}$$

$$\boxed{\text{z538}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{6}\left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g)\right)}{6}$$

$$\boxed{\text{z539}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{2u}) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g})}{6}$$

$$\boxed{\text{z540}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{2u}) = \frac{-2\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g})}{6}$$

$$\boxed{\text{z541}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{2u}) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z542}} \quad \mathbb{G}_{2,1}^{(c)}(T_{2u}) = \frac{\sqrt{2}\left(-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g})\right)}{6}$$

$$\begin{aligned}
\boxed{\text{z543}} \quad \mathbb{G}_{2,2}^{(c)}(T_{2u}) &= \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{6} \\
\boxed{\text{z544}} \quad \mathbb{G}_{2,3}^{(c)}(T_{2u}) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{1,1}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{Q}_{1,2}^{(a)}(T_{1u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 2\mathbb{Q}_{1,3}^{(a)}(T_{1u})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6} \\
\boxed{\text{z545}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}, a) &= \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z546}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}, a) &= \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z547}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}, a) &= \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z548}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}, b) &= \frac{\sqrt{14} \left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{42} \\
\boxed{\text{z549}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}, b) &= \frac{\sqrt{14} \left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{42} \\
\boxed{\text{z550}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}, b) &= \frac{\sqrt{14} \left(-2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42} \\
\boxed{\text{z551}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{14} \left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{84} \\
\boxed{\text{z552}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{14} \left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(E_g) + 4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{84} \\
\boxed{\text{z553}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_{2u}) &= \frac{\sqrt{14} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42}
\end{aligned}$$

$$\boxed{\text{z554}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_{2u}) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{6}$$

$$\boxed{\text{z555}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_{2u}) = \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3} \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{6}$$

$$\boxed{\text{z556}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_{2u}) = \frac{\sqrt{2} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 2 \mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u}) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z557}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_{2u}) = \frac{\sqrt{3} \mathbb{G}_0^{(1,1;a)}(A_{1u}) \mathbb{Q}_{2,1}^{(b)}(T_{2g})}{3}$$

$$\boxed{\text{z558}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_{2u}) = \frac{\sqrt{3} \mathbb{G}_0^{(1,1;a)}(A_{1u}) \mathbb{Q}_{2,2}^{(b)}(T_{2g})}{3}$$

$$\boxed{\text{z559}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_{2u}) = \frac{\sqrt{3} \mathbb{G}_0^{(1,1;a)}(A_{1u}) \mathbb{Q}_{2,3}^{(b)}(T_{2g})}{3}$$

• **A;A_002_1** : 'A'-'A' bond-cluster

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: **6b@3c**

$$\boxed{\text{z15}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z16}} \quad \mathbb{Q}_0^{(c)}(A_{1g}, b) = \frac{\sqrt{5} \left(\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(a)}(T_{2g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{5}$$

$$\boxed{\text{z17}} \quad \mathbb{Q}_4^{(c)}(A_{1g}) = \frac{\sqrt{30} \left(3 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2 \mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2 \mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2 \mathbb{Q}_{2,3}^{(a)}(T_{2g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z18}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_{1g}) = \frac{\sqrt{5} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{5}$$

$$\boxed{\text{z19}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_{1g}) = \frac{\sqrt{30} \left(3 \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2 \mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3 \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2 \mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2 \mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z20}} \quad \mathbb{Q}_0^{(1,1;c)}(A_{1g}) = \mathbb{Q}_0^{(1,1;a)}(A_{1g}) \mathbb{Q}_0^{(b)}(A_{1g})$$

$$\begin{aligned}
\boxed{\text{z33}} \quad \mathbb{G}_0^{(c)}(A_{1u}) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3} \\
\boxed{\text{z34}} \quad \mathbb{G}_0^{(1,-1;c)}(A_{1u}) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3} \\
\boxed{\text{z35}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, a) &= \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3} \\
\boxed{\text{z36}} \quad \mathbb{G}_4^{(1,-1;c)}(A_{1u}, b) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3} \\
\boxed{\text{z37}} \quad \mathbb{G}_0^{(1,0;c)}(A_{1u}) &= \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3} \\
\boxed{\text{z38}} \quad \mathbb{G}_0^{(1,1;c)}(A_{1u}) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3} \\
\boxed{\text{z45}} \quad \mathbb{G}_3^{(c)}(A_{2g}) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2} \\
\boxed{\text{z46}} \quad \mathbb{G}_3^{(1,-1;c)}(A_{2g}) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2} \\
\boxed{\text{z47}} \quad \mathbb{G}_3^{(1,0;c)}(A_{2g}) &= \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{3} \\
\boxed{\text{z55}} \quad \mathbb{Q}_3^{(c)}(A_{2u}) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3} \\
\boxed{\text{z56}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}, a) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3} \\
\boxed{\text{z57}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}, b) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3} \\
\boxed{\text{z58}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_{2u}, c) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3}
\end{aligned}$$

$$\boxed{\text{z59}} \quad \mathbb{Q}_3^{(1,0;c)}(A_{2u}) = \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{3}$$

$$\boxed{\text{z60}} \quad \mathbb{Q}_3^{(1,1;c)}(A_{2u}) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{3}$$

$$\boxed{\text{z99}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z100}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z101}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z102}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z103}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\sqrt{7} \left(2\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{14}$$

$$\boxed{\text{z104}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = \frac{\sqrt{7} \left(-2\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{14}$$

$$\boxed{\text{z105}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g) = \frac{\sqrt{21} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{42}$$

$$\boxed{\text{z106}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g) = \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{14}$$

$$\boxed{\text{z107}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z108}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$$\boxed{\text{z109}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{7} \left(2\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{14}$$

$$\begin{aligned}
\text{z110} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{7} \left(-2\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{14} \\
\text{z111} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g) &= \frac{\sqrt{21} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{42} \\
\text{z112} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{14} \\
\text{z113} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g) &= \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g})}{2} \\
\text{z114} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g) &= \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 2\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) \right)}{6} \\
\text{z115} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g) &= \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} \\
\text{z116} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g) &= \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} \\
\text{z155} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u) &= \frac{\sqrt{14} \left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{28} \\
\text{z156} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u) &= \frac{\sqrt{14} \left(\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{84} \\
\text{z157} \quad \mathbb{G}_{2,1}^{(c)}(E_u, a) &= \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6} \\
\text{z158} \quad \mathbb{G}_{2,2}^{(c)}(E_u, a) &= \frac{\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2} \\
\text{z159} \quad \mathbb{G}_{2,1}^{(c)}(E_u, b) &= \frac{\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u})}{2} \\
\text{z160} \quad \mathbb{G}_{2,2}^{(c)}(E_u, b) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\text{z161} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) &= \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{28} \\
\text{z162} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) &= \frac{\sqrt{14} \left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{84} \\
\text{z163} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) &= \frac{\sqrt{14} \left(-9\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{84} \\
\text{z164} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) &= \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{28} \\
\text{z165} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, c) &= \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6} \\
\text{z166} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, c) &= \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2} \\
\text{z167} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, d) &= \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u})}{2} \\
\text{z168} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, d) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{6} \\
\text{z169} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u) &= \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{84} \\
\text{z170} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u) &= \frac{\sqrt{14} \left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{28} \\
\text{z171} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, a) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{2} \\
\text{z172} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, a) &= \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z173}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, b) = \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z174}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, b) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u})}{2}$$

$$\boxed{\text{z175}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, a) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z176}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, a) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u})}{2}$$

$$\boxed{\text{z177}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) - \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u})}{2}$$

$$\boxed{\text{z178}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, b) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z236}} \quad \mathbb{Q}_{4,1}^{(c)}(T_{1g}) = -\frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{12}$$

$$\boxed{\text{z237}} \quad \mathbb{Q}_{4,2}^{(c)}(T_{1g}) = \frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\boxed{\text{z238}} \quad \mathbb{Q}_{4,3}^{(c)}(T_{1g}) = \frac{\sqrt{6} \left(\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z239}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_{1g}) = -\frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{12}$$

$$\boxed{\text{z240}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\boxed{\text{z241}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{6} \left(\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z242}} \quad \mathbb{G}_{1,1}^{(c)}(T_{1g}) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z243}} \quad \mathbb{G}_{1,2}^{(c)}(T_{1g}) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z244}} \quad \mathbb{G}_{1,3}^{(c)}(T_{1g}) = \frac{\sqrt{30} \left(-\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 2\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z245}} \quad \mathbb{G}_{3,1}^{(c)}(T_{1g}) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{60}$$

$$\boxed{\text{z246}} \quad \mathbb{G}_{3,2}^{(c)}(T_{1g}) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{60}$$

$$\boxed{\text{z247}} \quad \mathbb{G}_{3,3}^{(c)}(T_{1g}) = \frac{\sqrt{30} \left(-2\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 2\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z248}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z249}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30}$$

$$\boxed{\text{z250}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{30} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z251}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{60}$$

$$\boxed{\text{z252}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{60}$$

$$\boxed{\text{z253}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_{1g}) = \frac{\sqrt{30} \left(-2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\begin{aligned}
\boxed{\text{z254}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_{1g}, a) &= \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z255}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_{1g}, a) &= \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z256}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_{1g}, a) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_0^{(b)}(A_{1g})}{3} \\
\boxed{\text{z257}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_{1g}, b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30} \\
\boxed{\text{z258}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_{1g}, b) &= \frac{\sqrt{10} \left(3\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30} \\
\boxed{\text{z259}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_{1g}, b) &= \frac{\sqrt{10} \left(3\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{30} \\
\boxed{\text{z260}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_{1g}) &= \frac{\sqrt{5} \left(-3\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{30} \\
\boxed{\text{z261}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_{1g}) &= -\frac{\sqrt{5} \left(2\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{30} \\
\boxed{\text{z262}} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_{1g}) &= \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{15} \\
\boxed{\text{z341}} \quad \mathbb{Q}_{1,1}^{(c)}(T_{1u}, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z342}} \quad \mathbb{Q}_{1,2}^{(c)}(T_{1u}, a) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z343}} \quad \mathbb{Q}_{1,3}^{(c)}(T_{1u}, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\boxed{\text{z344}} \quad \mathbb{Q}_{1,1}^{(c)}(T_{1u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z345}} \quad \mathbb{Q}_{1,2}^{(c)}(T_{1u}, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z346}} \quad \mathbb{Q}_{1,3}^{(c)}(T_{1u}, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z347}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{21}$$

$$\boxed{\text{z348}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) - \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{21}$$

$$\boxed{\text{z349}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{21}$$

$$\boxed{\text{z350}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z351}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}, b) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z352}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z353}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_{1u}, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z354}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_{1u}, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z355}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_{1u}, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\begin{aligned}
\text{z356} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{1u}, a) &= \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{12} \\
\text{z357} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{1u}, a) &= \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{12} \\
\text{z358} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{1u}, a) &= \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{12} \\
\text{z359} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{1u}, b) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - 5\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 5\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 16\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{84} \\
\text{z360} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{1u}, b) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + 5\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - 5\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - 16\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{84} \\
\text{z361} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{1u}, b) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 5\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + 5\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - 16\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) \right)}{84} \\
\text{z362} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(T_{1u}, 2) &= \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u})}{12} \\
\text{z363} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(T_{1u}, 2) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u})}{12} \\
\text{z364} \quad \mathbb{Q}_{5,3}^{(1,-1;c)}(T_{1u}, 2) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u})}{12} \\
\text{z365} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_{1u}, a) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{30} \\
\text{z366} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_{1u}, a) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30} \\
\text{z367} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_{1u}, a) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30} \\
\text{z368} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_{1u}, b) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\text{z369} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_{1u}, b) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z370} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_{1u}, b) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + 2\mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z371} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{1u}, a) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + 3\sqrt{3} \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - 2\sqrt{3} \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - 2\sqrt{3} \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{30} \\
\text{z372} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{1u}, a) &= -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + 3\sqrt{3} \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\sqrt{3} \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{30} \\
\text{z373} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{1u}, a) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{3} \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{15} \\
\text{z374} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{1u}, b) &= \frac{\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{M}_{2,1}^{(b)}(T_{2u}) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{M}_{2,1}^{(b)}(T_{2u}) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u})}{6} \\
\text{z375} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{1u}, b) &= \frac{-\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u})}{6} \\
\text{z376} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{1u}, b) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u})}{3} \\
\text{z377} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_{1u}, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z378} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_{1u}, a) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z379} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_{1u}, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z380} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_{1u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6} \\
\text{z381} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_{1u}, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z382}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_{1u}, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z383}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z384}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_{1u}) = \frac{-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z385}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_{1u}) = \frac{\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z449}} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,1}^{(b)}(T_{2g})}{3}$$

$$\boxed{\text{z450}} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g})}{3}$$

$$\boxed{\text{z451}} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_{1g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g})}{3}$$

$$\boxed{\text{z452}} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}, b) = \frac{\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z453}} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}, b) = \frac{\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z454}} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}, b) = \frac{\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_{2g}) \mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z455}} \quad \mathbb{Q}_{2,1}^{(c)}(T_{2g}, c) = \frac{\sqrt{14} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3 \mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) - 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3 \mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3 \mathbb{Q}_{2,3}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{42}$$

$$\boxed{\text{z456}} \quad \mathbb{Q}_{2,2}^{(c)}(T_{2g}, c) = \frac{\sqrt{14} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3 \mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3 \mathbb{Q}_{2,3}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{42}$$

$$\boxed{\text{z457}} \quad \mathbb{Q}_{2,3}^{(c)}(T_{2g}, c) = \frac{\sqrt{14} \left(-2\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3 \mathbb{Q}_{2,1}^{(a)}(T_{2g}) \mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3 \mathbb{Q}_{2,2}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_{2g}) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42}$$

$$\boxed{\text{z458}} \quad \mathbb{Q}_{4,1}^{(c)}(T_{2g}) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{84}$$

$$\boxed{\text{z459}} \quad \mathbb{Q}_{4,2}^{(c)}(T_{2g}) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{84}$$

$$\boxed{\text{z460}} \quad \mathbb{Q}_{4,3}^{(c)}(T_{2g}) = \frac{\sqrt{14} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42}$$

$$\boxed{\text{z461}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z462}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z463}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_{2g}, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_0^{(b)}(A_{1g})}{3}$$

$$\boxed{\text{z464}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_{2g}, b) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{42}$$

$$\boxed{\text{z465}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_{2g}, b) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{42}$$

$$\boxed{\text{z466}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_{2g}, b) = \frac{\sqrt{14} \left(-2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42}$$

$$\boxed{\text{z467}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_{2g}) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{84}$$

$$\boxed{\text{z468}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_{2g}) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{84}$$

$$\begin{aligned}
\text{z469} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{14} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42} \\
\text{z470} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_{2g}) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) \right)}{6} \\
\text{z471} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_{2g}) &= \frac{\sqrt{2} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{6} \\
\text{z472} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_{2g}) &= \frac{\sqrt{2} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6} \\
\text{z473} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_{2g}) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g})}{3} \\
\text{z474} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_{2g}) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g})}{3} \\
\text{z475} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_{2g}) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g})}{3} \\
\text{z476} \quad \mathbb{G}_{3,1}^{(c)}(T_{2g}) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{12} \\
\text{z477} \quad \mathbb{G}_{3,2}^{(c)}(T_{2g}) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12} \\
\text{z478} \quad \mathbb{G}_{3,3}^{(c)}(T_{2g}) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{6} \\
\text{z479} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_{2g}) \right)}{12} \\
\text{z480} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12} \\
\text{z481} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_{2g}) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g})\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z482}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_{2g}) = \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) - 2\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g})}{6}$$

$$\boxed{\text{z483}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_{2g}) = \frac{-2\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,3}^{(b)}(T_{2g}) + \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g})}{6}$$

$$\boxed{\text{z484}} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_{2g}) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(T_{2g}) - \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,1}^{(b)}(T_{2g}) + \mathbb{G}_{1,3}^{(1,0;a)}(T_{1g})\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z560}} \quad \mathbb{Q}_{3,1}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z561}} \quad \mathbb{Q}_{3,2}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z562}} \quad \mathbb{Q}_{3,3}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \mathbb{M}_{1,2}^{(a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z563}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{2u}, a) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 4\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z564}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{2u}, a) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 4\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z565}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{2u}, a) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u}) + 4\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u})}{12}$$

$$\boxed{\text{z566}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{2u}, b) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - 3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{24}$$

$$\boxed{\text{z567}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{2u}, b) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - 3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - 3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{24}$$

$$\boxed{\text{z568}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{2u}, b) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - 3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - 3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{24}$$

$$\boxed{\text{z569}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_{2u}, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z570}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_{2u},c) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z571}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_{2u},c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z572}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{2} \left(-3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + 3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{24}$$

$$\boxed{\text{z573}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{2} \left(3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - 3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{24}$$

$$\boxed{\text{z574}} \quad \mathbb{Q}_{5,3}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{2} \left(-3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{24}$$

$$\boxed{\text{z575}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{2u},a) = \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_{1u}) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u})}{6}$$

$$\boxed{\text{z576}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{2u},a) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,3}^{(b)}(T_{1u}) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{6}$$

$$\boxed{\text{z577}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{2u},a) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g})\mathbb{T}_{1,1}^{(b)}(T_{1u})}{3}$$

$$\boxed{\text{z578}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_{2u},b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{2,1}^{(b)}(T_{2u}) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{2,1}^{(b)}(T_{2u})}{6}$$

$$\boxed{\text{z579}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_{2u},b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{2,2}^{(b)}(T_{2u}) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{2,2}^{(b)}(T_{2u})}{6}$$

$$\boxed{\text{z580}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_{2u},b) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{2,3}^{(b)}(T_{2u})}{3}$$

$$\boxed{\text{z581}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) - \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z582}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(T_{2u}) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g})\mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g})\mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z583}} \quad \mathbb{Q}_{3,3}^{(1,1;c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) - \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6}$$

$$\boxed{\text{z584}} \quad \mathbb{G}_{2,1}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z585}} \quad \mathbb{G}_{2,2}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z586}} \quad \mathbb{G}_{2,3}^{(c)}(T_{2u}) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z587}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{21}$$

$$\boxed{\text{z588}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{21}$$

$$\boxed{\text{z589}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{21}$$

$$\boxed{\text{z590}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_{2u}, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z591}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_{2u}, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z592}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_{2u}, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}$$

$$\boxed{\text{z593}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + 12 \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{84}$$

$$\boxed{\text{z594}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + 12 \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{84}$$

$$\boxed{\text{z595}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_{2u}) = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + 12 \mathbb{M}_3^{(1,-1;a)}(A_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) \right)}{84}$$

$$\begin{aligned}
\text{z596} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_{2u}, a) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_{1u}) - \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z597} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_{2u}, a) &= \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z598} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_{2u}, a) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) - 2 \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z599} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_{2u}, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) \right)}{6} \\
\text{z600} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_{2u}, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,3}^{(b)}(T_{2u}) + \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z601} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_{2u}, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,2}^{(b)}(T_{2u}) + \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) \mathbb{M}_{2,1}^{(b)}(T_{2u}) \right)}{6} \\
\text{z602} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) \right)}{6} \\
\text{z603} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,3}^{(b)}(T_{1u}) + \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6} \\
\text{z604} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_{2u}) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,2}^{(b)}(T_{1u}) + \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) \mathbb{T}_{1,1}^{(b)}(T_{1u}) \right)}{6}
\end{aligned}$$

Atomic SAMB

- bra: $\langle s, \uparrow |, \langle s, \downarrow |$
- ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$

$$\text{x1} \quad \mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$\boxed{\text{x2}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x3}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x4}} \quad \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

- bra: $\langle s, \uparrow |, \langle s, \downarrow |$
- ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

$$\boxed{\text{x5}} \quad \mathbb{Q}_{1,1}^{(a)}(T_{1u}) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x6}} \quad \mathbb{Q}_{1,2}^{(a)}(T_{1u}) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x7}} \quad \mathbb{Q}_{1,3}^{(a)}(T_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$$\boxed{\text{x8}} \quad \mathbb{Q}_{1,1}^{(1,0;a)}(T_{1u}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x9}} \quad \mathbb{Q}_{1,2}^{(1,0;a)}(T_{1u}) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x10}} \quad \mathbb{Q}_{1,3}^{(1,0;a)}(T_{1u}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x11}} \quad \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$$

$$\boxed{\text{x12}} \quad \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x13}} \quad \mathbb{G}_{2,1}^{(1,-1;a)}(T_{2u}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x14}} \quad \mathbb{G}_{2,2}^{(1,-1;a)}(T_{2u}) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x15}} \quad \mathbb{G}_{2,3}^{(1,-1;a)}(T_{2u}) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x16}} \quad \mathbb{G}_0^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$$

$$\boxed{\text{x17}} \quad \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{6} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x18}} \quad \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x19}} \quad \mathbb{M}_{2,1}^{(1,-1;a)}(T_{2u}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x20}} \quad \mathbb{M}_{2,2}^{(1,-1;a)}(T_{2u}) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x21}} \quad \mathbb{M}_{2,3}^{(1,-1;a)}(T_{2u}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x22}} \quad \mathbb{M}_0^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$$

$$\boxed{\text{x23}} \quad \mathbb{T}_{1,1}^{(a)}(T_{1u}) = \begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x24}} \quad \mathbb{T}_{1,2}^{(a)}(T_{1u}) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x25}} \quad \mathbb{T}_{1,3}^{(a)}(T_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$$

$$\boxed{\text{x26}} \quad \mathbb{T}_{1,1}^{(1,0;a)}(T_{1u}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x27}} \quad \mathbb{T}_{1,2}^{(1,0;a)}(T_{1u}) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x28}} \quad \mathbb{T}_{1,3}^{(1,0;a)}(T_{1u}) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

- bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$
- ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

$$\boxed{\text{x29}} \quad \mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x30}} \quad \mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$$

$$\boxed{\text{x31}} \quad \mathbb{Q}_{2,2}^{(a)}(E_g) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x32}} \quad \mathbb{Q}_{2,1}^{(a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x33}} \quad \mathbb{Q}_{2,2}^{(a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x34}} \quad \mathbb{Q}_{2,3}^{(a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x35}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x36}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x37}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x38}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x39}} \quad \mathbb{Q}_{2,3}^{(1,-1;a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x40}} \quad \mathbb{Q}_0^{(1,1;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x41}} \quad \mathbb{G}_{1,1}^{(1,0;a)}(T_{1g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x42}} \quad \mathbb{G}_{1,2}^{(1,0;a)}(T_{1g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x43}} \quad \mathbb{G}_{1,3}^{(1,0;a)}(T_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x44}} \quad \mathbb{M}_{1,1}^{(a)}(T_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x45}} \quad \mathbb{M}_{1,2}^{(a)}(T_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x46}} \quad \mathbb{M}_{1,3}^{(a)}(T_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x47}} \quad \mathbb{M}_3^{(1,-1;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x48}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x49}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x50}} \quad \mathbb{M}_{1,3}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x51}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{5}}{5} & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{5}}{5} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x52}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{5} & -\frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x53}} \quad \mathbb{M}_{3,3}^{(1,-1;a)}(T_{1g}) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$$

$$\boxed{\text{x54}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x55}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(T_{2g}) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x56}} \quad \mathbb{M}_{3,3}^{(1,-1;a)}(T_{2g}) = \begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x57}} \quad \mathbb{M}_{1,1}^{(1,1;a)}(T_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$$

$$\boxed{\text{x58}} \quad \mathbb{M}_{1,2}^{(1,1;a)}(T_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$$

$$\boxed{\text{x59}} \quad \mathbb{M}_{1,3}^{(1,1;a)}(T_{1g}) = \begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$$

$$\boxed{\text{x60}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x61}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x62}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(T_{2g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x63}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(T_{2g}) = \begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x64}} \quad \mathbb{T}_{2,3}^{(1,0;a)}(T_{2g}) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$$

Cluster SAMB

- Site cluster

** Wyckoff: 1a

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_{1g}) = [1]$$

- Bond cluster

** Wyckoff: 3a@3d

$$\boxed{\text{y2}} \quad \mathbb{Q}_0^{(b)}(A_{1g}) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y3}} \quad \mathbb{Q}_{2,1}^{(b)}(E_g) = \left[-\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3} \right]$$

$$\boxed{\text{y4}} \quad \mathbb{Q}_{2,2}^{(b)}(E_g) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0 \right]$$

$$\boxed{\text{y5}} \quad \mathbb{T}_{1,1}^{(b)}(T_{1u}) = [i, 0, 0]$$

$$\boxed{\text{y6}} \quad \mathbb{T}_{1,2}^{(b)}(T_{1u}) = [0, i, 0]$$

$$\boxed{\text{y7}} \quad \mathbb{T}_{1,3}^{(b)}(T_{1u}) = [0, 0, i]$$

** Wyckoff: **6b@3c**

$$\boxed{\text{y8}} \quad \mathbb{Q}_0^{(b)}(A_{1g}) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y9}} \quad \mathbb{Q}_{2,1}^{(b)}(E_g) = \left[-\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y10}} \quad \mathbb{Q}_{2,2}^{(b)}(E_g) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, 0, 0 \right]$$

$$\boxed{\text{y11}} \quad \mathbb{T}_{1,1}^{(b)}(T_{1u}) = \left[0, 0, \frac{i}{2}, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2} \right]$$

$$\boxed{\text{y12}} \quad \mathbb{T}_{1,2}^{(b)}(T_{1u}) = \left[\frac{i}{2}, -\frac{i}{2}, 0, 0, \frac{i}{2}, \frac{i}{2} \right]$$

$$\boxed{\text{y13}} \quad \mathbb{T}_{1,3}^{(b)}(T_{1u}) = \left[\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, 0, 0 \right]$$

$$\boxed{\text{y14}} \quad \mathbb{Q}_{2,1}^{(b)}(T_{2g}) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y15}} \quad \mathbb{Q}_{2,2}^{(b)}(T_{2g}) = \left[0, 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0, 0 \right]$$

$$\boxed{\text{y16}} \quad \mathbb{Q}_{2,3}^{(b)}(T_{2g}) = \left[0, 0, 0, 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

$$\boxed{\text{y17}} \quad \mathbb{M}_{2,1}^{(b)}(T_{2u}) = \left[0, 0, \frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, \frac{i}{2} \right]$$

$$\boxed{\text{y18}} \quad \mathbb{M}_{2,2}^{(b)}(T_{2u}) = \left[-\frac{i}{2}, \frac{i}{2}, 0, 0, \frac{i}{2}, \frac{i}{2} \right]$$

$$\boxed{\text{y19}} \quad \mathbb{M}_{2,3}^{(b)}(T_{2u}) = \left[\frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, 0, 0 \right]$$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	A	$ s, \uparrow\rangle, s, \downarrow\rangle, p_x, \uparrow\rangle, p_x, \downarrow\rangle, p_y, \uparrow\rangle, p_y, \downarrow\rangle, p_z, \uparrow\rangle, p_z, \downarrow\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	A	A	[1, 2]	[s, p]	[s, p]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'A' (#1) site cluster (1a), m-3m

SL	position (s)	mapping
1	[0.00000, 0.00000, 0.00000]	[1,2,3,4,...,48]

— Bond in Unit Cell —

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'A'-'A' [1] (#1) bond cluster (3a@3d), ND, $|v|=1.0$ (cartesian)

SL	vector (v)	center (c)	mapping	head	tail	R (primitive)
1	[-1.00000, 0.00000, 0.00000]	[0.50000, 0.00000, 0.00000]	[1,-2,-3,4,17,-18,-19,20,-25,26,27,-28,-41,42,43,-44]	(1,1)	(1,1)	[1,0,0]
2	[0.00000,-1.00000, 0.00000]	[0.00000, 0.50000, 0.00000]	[5,-6,-7,8,13,-14,-15,16,-29,30,31,-32,-37,38,39,-40]	(1,1)	(1,1)	[0,1,0]
3	[0.00000, 0.00000,-1.00000]	[0.00000, 0.00000, 0.50000]	[9,-10,-11,12,-21,22,23,-24,-33,34,35,-36,45,-46,-47,48]	(1,1)	(1,1)	[0,0,1]

Table 9: 2-th 'A'-'A' [1] (#2) bond cluster (6b@3c), ND, $|v|=1.41421$ (cartesian)

SL	vector (v)	center (c)	mapping	head	tail	R (primitive)
1	[0.00000,-1.00000,-1.00000]	[0.00000, 0.50000, 0.50000]	[1,-4,18,-19,-25,28,-42,43]	(1,1)	(1,1)	[0,1,1]
2	[0.00000, 1.00000,-1.00000]	[0.00000, 0.50000, 0.50000]	[2,-3,-17,20,-26,27,41,-44]	(1,1)	(1,1)	[0,-1,1]

continued ...

Table 9

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
3	$[-1.00000, 0.00000, -1.00000]$	$[0.50000, 0.00000, 0.50000]$	$[5, -8, -14, 15, -29, 32, 38, -39]$	$(1, 1)$	$(1, 1)$	$[1, 0, 1]$
4	$[-1.00000, 0.00000, 1.00000]$	$[0.50000, 0.00000, 0.50000]$	$[6, -7, 13, -16, -30, 31, -37, 40]$	$(1, 1)$	$(1, 1)$	$[1, 0, -1]$
5	$[-1.00000, -1.00000, 0.00000]$	$[0.50000, 0.50000, 0.00000]$	$[9, -12, 21, -24, -33, 36, -45, 48]$	$(1, 1)$	$(1, 1)$	$[1, 1, 0]$
6	$[1.00000, -1.00000, 0.00000]$	$[0.50000, 0.50000, 0.00000]$	$[10, -11, -22, 23, -34, 35, 46, -47]$	$(1, 1)$	$(1, 1)$	$[-1, 1, 0]$